

Customer Churn Prediction Pipeline

An end-to-end machine learning approach to identifying customers likely to churn, using demographic, behavioral, and engagement data.

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Objective & Approach

Predicting which customers are likely to churn, so marketing can act before they leave

01

Explore & Clean

Inspect structure, missing values, and distributions across two datasets.

02

Preprocess

Impute missing values; scale numeric features; one-hot encode categoricals.

03

Train Model

Fit a Random Forest classifier on an 80/20 stratified train-test split.

04

Evaluate

Score with accuracy, precision, recall, F1, and ROC-AUC; inspect the confusion matrix.

05

Interpret & Act

Rank feature importances to surface the strongest drivers of churn.

Data & Methodology

Two customer datasets, one consistent preprocessing and modeling pipeline

Dataset 1

40,000 rows

51.8% churn rate · 14 columns · no missing values

Dataset 2

50,000 rows

87.3% churn rate · 14 columns · same schema as Dataset 1

Features used

age, gender, region, income, tenure, monthly spend, website visits, support tickets, email click rate, loyalty score, campaign type, discount used, days since last campaign

Pipeline Configuration

- **Split**

80% train / 20% test, stratified by churn (random_state = 42)

- **Numeric features**

Median imputation → StandardScaler

- **Categorical features**

Most-frequent imputation → One-Hot Encoding

- **Model**

Random Forest Classifier (200 trees)

- **Evaluation**

Accuracy, Precision, Recall, F1, ROC-AUC, Confusion Matrix

Model Performance

Random Forest evaluated on the held-out 20% test split (Dataset 2)

94.0%

Accuracy

95.3%

Precision

98.0%

Recall

96.6%

F1 Score

0.964

ROC-AUC

Confusion Matrix

851

True Negative

421

False Positive

178

False Negative

8,550

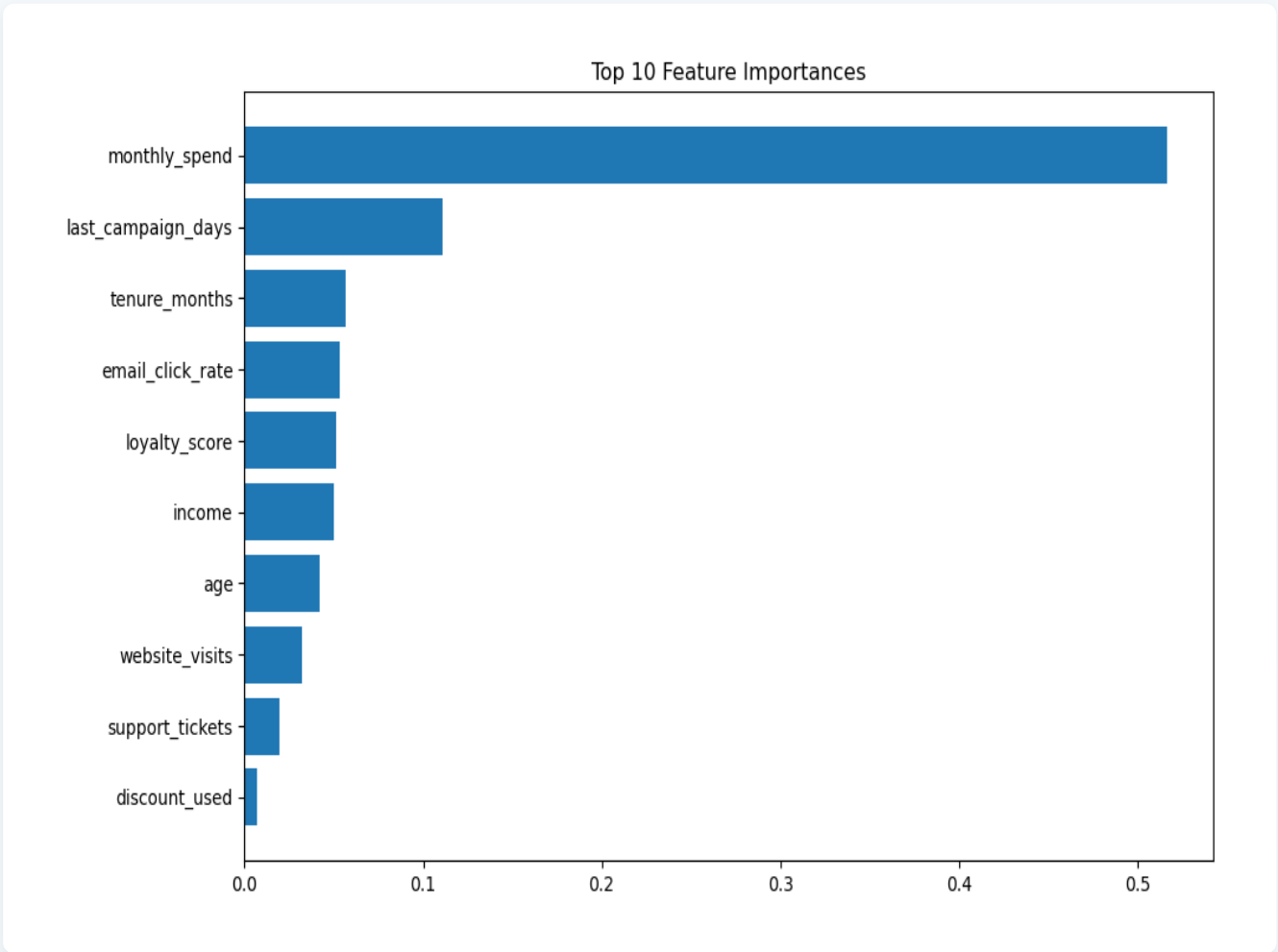
True Positive

Key Takeaways

- The model correctly flags 98% of customers who actually churn (high recall) — critical for a retention use case where missed churners are costly.
- Precision of 95.3% means retention outreach is well-targeted, with a relatively small share of false alarms.
- ROC-AUC of 0.96 indicates strong separation between churners and non-churners across thresholds.

Feature Importance & Business Insights

Which signals drive the model's churn predictions



Top Drivers

1

Monthly Spend

By far the strongest predictor — spending patterns signal disengagement early.

2

Days Since Last Campaign

Longer gaps since outreach correlate with higher churn risk.

3

Tenure & Email Engagement

Newer, less-engaged customers are more likely to leave.

Recommendation

Prioritize retention offers for customers with declining spend and long gaps since their last campaign touchpoint — the model flags these as the clearest early-warning signals.